

1. (a) $f(x) = x - \cos x \Rightarrow f'(x) = 1 + \sin x$; $f''(x) = \cos x$; $f^{(3)}(x) = -\sin x$; ...

$$\begin{aligned} P_n(x) &= f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \dots + \frac{f^{(n)}(0)}{n!}x^n \\ &= -1 + x + \frac{x^2}{2} - \frac{x^4}{4!} + \frac{x^6}{6!} - \dots + \frac{f^{(n)}(0)}{n!}x^n \\ &= x - \sum_{k=0}^n \frac{(-1)^k}{(2k)!}x^{2k} \end{aligned}$$

(b) Let $t = rx \Rightarrow P_n(x) = e^{rx} = e^t = \sum_{k=0}^n \frac{t^k}{k!} = \sum_{k=0}^n \frac{a^k}{k!}x^k$

- (c) Since $f(x) = \sinh x$, then

$$\frac{d^k}{dx^k} \sinh x = \begin{cases} \sinh x, & \text{if } k \text{ is odd} \\ \cosh x, & \text{if } k \text{ is even} \end{cases}$$

Thus

$$\frac{d^k}{dx^k} \sinh x|_{x=0} = \begin{cases} 0, & \text{if } k \text{ is odd} \\ 1, & \text{if } k \text{ is even} \end{cases}$$

and

$$P_n(x) = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots + \frac{f^{(n)}(0)}{n!}x^n = \sum_{k=0}^n \frac{x^{2k+1}}{(2k+1)!}$$

- (d) $f(x) = \ln(1-x) \Rightarrow f'(x) = \frac{-1}{1-x}$; $f''(x) = \frac{-1}{(1-x)^2}$; $f^{(3)}(x) = \frac{-2}{(1-x)^3}$; ...

$$P_n(x) = -x + \frac{-1}{2!}x^2 + \frac{-2}{3!}x^3 + \dots + \frac{f^{(n)}(0)}{n!}x^n = -\sum_{k=1}^n \frac{1}{k}x^k$$

2. • Let $g(x) = \frac{x^{-n}}{e^{1/x^2}}$. Then $\lim_{x \rightarrow 0} g(x)$ has the form ∞/∞ .

Successive application of *L'Hôpital's rule* will finally produce a quotient of the form $\frac{cx^k}{e^{1/x^2}}$, where k is a nonnegative integer and c is a constant. It follows that $\lim_{x \rightarrow 0} g(x) = 0$.

- $f'(0) = \lim_{x \rightarrow 0} \frac{e^{1/x^2} - 0}{x} = 0$ by previous problem.

Assume that $f^{(k)}(0) = 0$

$$\text{Then } f^{(k+1)}(0) = \lim_{x \rightarrow 0} \frac{f^{(k)}(x) - 0}{x} = \lim_{x \rightarrow 0} \frac{f^{(k)}(x)}{x}$$

Now, $\frac{f^{(k)}(x)}{x}$ is a sum of terms of the form $\frac{ce^{-1/x^2}}{x^n}$, n is a positive integer and c is a constant.

Again, by previous problem, $f^{(k+1)}(0) = 0$. Therefore $f^{(n)}(0) = 0$ for all n

- The Taylor series of $f(x)$ is 0.
- The Taylor series converges to $f(x)$ at $x = 0$.

3. (a) $f(x) = \arctan x$, $a = 1$, $n = 5$

$$\Rightarrow f'(x) = \frac{1}{1+x^2}; f''(x) = \frac{-2x}{(1+x^2)^2}; f^{(3)}(x) = \frac{6x^2-2}{(1+x^2)^3}; f^{(4)}(x) = \frac{24(x-x^3)}{(1+x^2)^4};$$

$$f^{(5)}(x) = \frac{24(1-10x^2+5x^4)}{(1+x^2)^5}; f^{(6)}(x) = \frac{-240x(3-10x^2+3x^4)}{(1+x^2)^6}$$

$$\text{and then } f'(1) = \frac{1}{2}; f(1) = \frac{\pi}{4}; f''(1) = \frac{1}{2}; f'''(1) = -\frac{1}{2}; f^{(4)}(1) = \frac{1}{2}; f^{(5)}(1) = -3$$

Therefore,

$$P_5(x) = \frac{\pi}{4} + \frac{1}{2}(x-1) - \frac{1/2}{2!}(x-1)^2 + \frac{1/2}{3!}(x-1)^3 + 0 + \frac{-3}{5!}(x-1)^5$$

$$= \frac{\pi}{4} + \frac{1}{2}(x-1) - \frac{1}{4}(x-1)^2 + \frac{1}{12}(x-1)^3 - \frac{1}{40}(x-1)^5$$

$$R_5(x) = \frac{f^{(6)}(c)}{6!}(x-1)^6 = \frac{-c(3-10c^2+3c^4)}{3(1+c^2)^6}(x-1)^6, \text{ where } c \text{ is between } 1 \text{ and } x.$$

- (b) $f(x) = \ln x$, $a = 1$, $n = 5$

$$\Rightarrow f'(x) = \frac{1}{x}; f''(x) = -\frac{1}{x^2}; f^{(3)}(x) = \frac{2}{x^3};$$

$$f^{(4)}(x) = -\frac{6}{x^4}; f^{(5)}(x) = \frac{24}{x^5}; f^{(6)}(x) = -\frac{120}{x^6}$$

$$\text{and then } f(1) = 0; f'(1) = 1; f''(1) = -1; f^{(3)}(1) = 2; f^{(4)}(1) = -6; f^{(5)}(1) = 24$$

Therefore,

$$P_5(x) = 0 + (x-1) - \frac{1}{2!}(x-1)^2 + \frac{2}{3!}(x-1)^3 - \frac{6}{4!}(x-1)^4 + \frac{24}{5!}(x-1)^5$$

$$= (x-1) - \frac{1}{2}(x-1)^2 + \frac{1}{3}(x-1)^3 - \frac{1}{4}(x-1)^4 + \frac{1}{5}(x-1)^5$$

$$R_5(x) = \frac{f^{(6)}(c)}{6!}(x-1)^6 = \frac{-120/c^6}{6!}(x-1)^6 = -\frac{1}{6}(x-1)^6, \text{ where } c \text{ is between } 1 \text{ and } x.$$

4. (a) $f(x) = \frac{1}{1-2x} = \frac{1}{5-2(x+2)} = \frac{1}{5} \left[\frac{1}{1+\frac{2}{5}(x+2)} \right]$

$$= \frac{1}{5} \sum_{k=0}^{\infty} \left[\frac{2}{5}(x+2) \right]^k = \sum_{k=0}^{\infty} \frac{2^k}{5^{k+1}}(x+2)^k$$

This result holds for $|\frac{2}{5}(x+2)| < 1$ and thus for $-\frac{9}{2} < x < \frac{1}{2}$

(b) $f(x) = \ln(1+2x) = \ln[3+2(x-1)] = \ln[2(1+\frac{2}{3}(x-1))]$

$$= \ln 3 + \ln \left[1 + \frac{2}{3}(x-1) \right] = \ln 3 + \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} \left[\frac{2}{3}(x-1) \right]^k$$

$$= \ln 3 + \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} \left(\frac{2}{3} \right)^k (x-1)^k$$

This result holds for $-1 < \frac{2}{3}(x-1) \leq 1$ and thus for $-\frac{1}{2} < x \leq \frac{5}{2}$

5. (a) Let $a_k = \left| \frac{x^k}{\sqrt{k}} \right| \Rightarrow \sqrt[k]{a_k} = \left| \frac{x^k}{\sqrt{k}} \right|^{\frac{1}{k}} = \frac{|x|}{k^{1/2}} \rightarrow |x|$ as $k \rightarrow \infty$

By Root test, $\sum \frac{(-1)^k}{\sqrt{k}} x^k$ converges if $|x| < 1$

If $x = 1$, $\sum \frac{(-1)^k}{\sqrt{k}} x^k$ converges; but if $x = -1$, $\sum \frac{1}{\sqrt{k}}$ diverges.
Therefore, the interval of convergence is $(-1, 1]$.

(b) Let $a_k = \left| \frac{2^k}{k^2} x^k \right| \Rightarrow \sqrt[k]{a_k} = \left| \frac{2^k}{k^2} x^k \right|^{\frac{1}{k}} = \frac{2|x|}{k^{2/k}} \rightarrow 2|x|$ as $k \rightarrow \infty$

By Root test, $\sum \frac{2^k}{k^2} x^k$ converges if $|x| < \frac{1}{2}$

If $x = \frac{1}{2}$, $\sum \frac{1}{k^2}$ converges; and if $x = -\frac{1}{2}$, $\sum \frac{(-1)^k}{k^2}$ converges.

Therefore, the interval of convergence is $\left[-\frac{1}{2}, \frac{1}{2}\right]$.

(c) Let $a_k = \frac{1}{\ln k} (x-1)^k \Rightarrow \sqrt[k]{a_k} = \left| \frac{1}{\ln k} (x-1)^k \right|^{\frac{1}{k}} = \frac{|x-1|}{\ln k} \rightarrow 0$ as $k \rightarrow \infty$

Therefore, by Root test, the interval of convergence is $(-\infty, \infty)$.

(d) Let $a_k = \frac{7^k}{k!} x^k \Rightarrow \frac{a_{k+1}}{a_k} = \frac{7^{k+1} |x^{k+1}|}{(k+1)! \cdot 7^k |x|^k} = \frac{7|x|}{k+1} \rightarrow 0$ as $k \rightarrow \infty$

Therefore, by Ratio test, the interval of convergence is $(-\infty, \infty)$.

(e) Let $a_k = \frac{k}{k-1} \left(\frac{x+2}{2}\right)^k \Rightarrow \frac{a_{k+1}}{a_k} = \frac{k^2-1}{2k^2} |x+2| \rightarrow \frac{|x+2|}{2}$ as $k \rightarrow \infty$

By Ratio test, $\sum \frac{k}{k-1} \left(\frac{x+2}{2}\right)^k$ converges if $|x+2| < 2$

But at the endpoints $x = 0$ or -4 , the series diverges since $\lim_{k \rightarrow \infty} a_k \neq 0$

Therefore, the interval of convergence is $(-4, 0)$.

(f) Let $a_k = \left(1 + \frac{1}{k}\right) x^k \Rightarrow \frac{a_{k+1}}{a_k} = \frac{1 + \frac{1}{k+1}}{1 + \frac{1}{k}} |x| = \left(1 - \frac{1}{(k+1)^2}\right) |x| \rightarrow |x|$ as $k \rightarrow \infty$

By Ratio test, $\sum \left(1 + \frac{1}{k}\right) x^k$ converges if $|x| < 1$

Therefore, the interval of convergence is $(-1, 1)$.

6. Since $\sum_{k=0}^{\infty} a_k x^k$ has the property $a_{k+3} = a_k$ for all $k \geq 0$,

then $\sum_{k=0}^{\infty} a_k x^k = a_0 + a_1 x + a_2 x^2 + a_0 x^3 + a_1 x^4 + a_2 x^5 + a_0 x^6 + \dots$

$$= \sum_{k=0}^{\infty} (a_0 + a_1 x + a_2 x^2) x^{3k} = \sum_{k=0}^{\infty} (a_0 + a_1 x + a_2 x^2) (x^3)^k$$

(a) Let $b_k = (a_0 + a_1 x + a_2 x^2) (x^3)^k \Rightarrow \frac{b_{k+1}}{b_k} = \left| \frac{(a_0 + a_1 x + a_2 x^2) (x^3)^{k+1}}{(a_0 + a_1 x + a_2 x^2) (x^3)^k} \right| = |x^3|$

Therefore, by Ratio test, the radius of convergence of the series $r = 1$.

(b) $\sum_{k=0}^{\infty} (a_0 + a_1 x + a_2 x^2) (x^3)^k = (a_0 + a_1 x + a_2 x^2) \sum_{k=0}^{\infty} (x^3)^k = (a_0 + a_1 x + a_2 x^2) \frac{1}{1-x^3}$

7. (a) Use the fact that $\frac{d}{dx} \left(\frac{1}{1-x} \right) = \frac{1}{(1-x)^2}$

Then $\frac{1}{(1-x)^2} = \frac{d}{dx} (1 + x + x^2 + x^3 + \dots + x^n + \dots) = 1 + 2x + 3x^2 + \dots + nx^{n-1} + \dots$

$$(b) \ln(2-3x) = \ln 2 + \ln\left(1 - \frac{3}{2}x\right) = \ln 2 - \frac{3}{2}x - \frac{1}{2}\left(\frac{3}{2}\right)^2 x^2 - \frac{1}{3}\left(\frac{3}{2}\right)^3 x^3 - \dots - \frac{1}{n+1}\left(\frac{3}{2}\right)^{n+1} x^{n+1} - \dots$$

$$8. (a) \sum_{k=0}^{\infty} \frac{1}{k!} x^{3k} = e^{3x}$$

$$(b) \sum_{k=0}^{\infty} \frac{1}{k!} x^{3k+1} = x \sum_{k=0}^{\infty} \frac{1}{k!} x^{3k} = x e^{3x}$$

$$(c) \sum_{k=0}^{\infty} \frac{3k}{k!} x^{3k-1} = \sum_{k=0}^{\infty} \frac{d}{dx} \frac{1}{k!} x^{3k} = \frac{d}{dx} (e^{x^3}) = 3x^2 e^{x^3}$$

$$9. (a) f(x) = x e^x = x \sum_{k=0}^{\infty} \frac{x^k}{k!} = \sum_{k=0}^{\infty} \frac{x^{k+1}}{k!}$$

$$(b) \text{ Using integration by part : } \int_0^1 x e^x dx = [x e^x - e^x]_0^1 = e - e + 1 = 1$$

Using the power series representation :

$$\begin{aligned} \int_0^1 x e^x dx &= \int_0^1 \left(\sum_{k=0}^{\infty} \frac{x^{k+1}}{k!} \right) dx = \sum_{k=0}^{\infty} \int_0^1 \left(\frac{x^{k+1}}{k!} \right) dx \\ &= \sum_{k=0}^{\infty} \frac{1}{k!} \left[\frac{x^{k+2}}{k+2} \right]_0^1 = \sum_{k=0}^{\infty} \frac{1}{k!(k+2)} = \frac{1}{2} + \sum_{k=1}^{\infty} \frac{1}{k!(k+2)} \end{aligned}$$

$$\text{Thus, } 1 = \frac{1}{2} + \sum_{k=1}^{\infty} \frac{1}{k!(k+2)} \Rightarrow \sum_{k=1}^{\infty} \frac{1}{k!(k+2)} = \frac{1}{2}.$$

$$10. \sum_{k=1}^{\infty} a_k = \sum_{k=1}^{\infty} \int_k^{k+1} x e^{-x} dx = \int_1^{\infty} x e^{-x} dx = \frac{2}{e}$$

$$11. (a) \sum_{k=0}^{\infty} \frac{(\ln 2)^k}{k!} = e^{\ln 2} = 2$$

$$(b) \sum_{k=0}^{\infty} \frac{3k}{k!} x^{3k-1} = \sum_{k=0}^{\infty} \frac{d}{dx} \frac{1}{k!} x^{3k} = \frac{d}{dx} (e^{x^3}) = 3x^2 e^{x^3}$$

$$\begin{aligned} 12. (a) \int_0^x \frac{\ln(1+t)}{t} dt &= \int_0^x \frac{1}{t} \left(\sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k} t^k \right) dt = \int_0^x \left(\sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k} t^{k-1} \right) dt \\ &= \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k} \int_0^x t^{k-1} dt = \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k^2} x^k, \text{ for } -1 \leq x \leq 1 \end{aligned}$$

$$\begin{aligned} (b) \int_0^x \frac{\arctan t}{t} dt &= \int_0^x \frac{1}{t} \left(\sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1} t^{2k+1} \right) dt = \int_0^x \left(\sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1} t^{2k} \right) dt \\ &= \sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1} \int_0^x t^{2k} dt = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)^2} x^{2k+1}, \text{ for } -1 \leq x \leq 1 \end{aligned}$$

$$13. (a) f(x) = \ln x = \ln[1 + (x-1)] = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} (x-1)^k$$

Radius of convergence : $r = 1$

$$(b) \quad f(x) = e^{-2x} = e^{-2(x+1)+2} = e^2 \cdot e^{-2(x+1)} = e^2 \sum_{k=0}^{\infty} \frac{[-2(x+1)]^k}{k!} = e^2 \sum_{k=0}^{\infty} \frac{(-1)^k 2^k}{k!} (x+1)^k$$

Radius of convergence : $r = \infty$

$$(c) \quad f(x) = \frac{2x}{1-x^2} = 2x \left(\frac{1}{1-x^2} \right) = 2x \sum_{k=0}^{\infty} (x^2)^k = \sum_{k=0}^{\infty} 2x^{2k+1}$$

Radius of convergence : $r = 1$

$$(d) \quad \text{Since } \sin x = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} x^{2k+1}$$

$$\text{Therefore, } f(x) = \sin x^2 = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} (x^2)^{2k+1} = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} x^{4k+2}$$

Radius of convergence : $r = \infty$

$$14. \quad (a) \quad \text{Since } \sum_{k=0}^{\infty} \left| \frac{(-1)^k (100)^k}{k!} \right| = \sum_{k=0}^{\infty} \frac{100^k}{k!} \text{ converges by the ratio test.}$$

Therefore, $\sum_{k=0}^{\infty} \frac{(-1)^k (100)^k}{k!}$ is absolutely convergent.

$$(b) \quad \text{Since } \sum_{k=1}^{\infty} \left| (-1)^{k-1} \frac{\ln k}{\sqrt{k}} \right| = \sum_{k=1}^{\infty} \frac{\ln k}{\sqrt{k}} \text{ diverges by the integral test.}$$

Therefore, $\sum_{k=1}^{\infty} (-1)^{k-1} \frac{\ln k}{\sqrt{k}}$ is conditionally convergent.

$$(c) \quad \text{Since } \sum_{k=1}^{\infty} \left| \frac{(-1)^k}{\sqrt{(k+1)(k+2)}} \right| = \sum_{k=1}^{\infty} \frac{1}{\sqrt{(k+1)(k+2)}} \text{ diverges}$$

by the limit comparison with $\sum_{k=1}^{\infty} \frac{1}{k}$.

Therefore, $\sum_{k=1}^{\infty} \frac{(-1)^k}{\sqrt{(k+1)(k+2)}}$ is conditionally convergent.